Electromagnetic Wave Optics

Part I: Mathematical Background and Basic Definitions

The following subjects will be covered

- Maxwell's Equations: Review
- The EM Wave Equation and Solutions.
- The Refractive Index Relation.
- Light Wave Power Density.

Summary

Consider the following Plane wave represented by E and H.

$$\vec{E} = E_o \cos(\omega t - kz)\, e^{-\alpha z}\ \vec{a}_x$$

$$\vec{H} = H_o \cos(\omega t - kz - \theta_\eta)\, e^{-\alpha z}\ \vec{a}_y$$

$$\vec{S} = \vec{E} \times \vec{H}$$

Poynting vector (Wave power density W/m^2)

$$\vec{S} = E_o H_o \cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

$$\vec{S} = \frac{{E_o}^2}{|\eta|}\cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

The time-average power density is given by

$$\vec{S}_{ave} = \frac{1}{T}\int_0^T \vec{S}\ dt\ \vec{a}_z$$

(W/m^2)

For a plane wave,

$$\vec{S}_{ave} = \frac{1}{2}\, Re\left[\vec{E} \times \vec{H}^*\right] \Rightarrow \vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|}\cos\theta_\eta\, e^{-2\alpha z}\ \vec{a}_z$$

Total average power within some area

$$P_{ave} = \int_A \vec{S} \cdot \overrightarrow{dA} \quad \text{(W)}$$

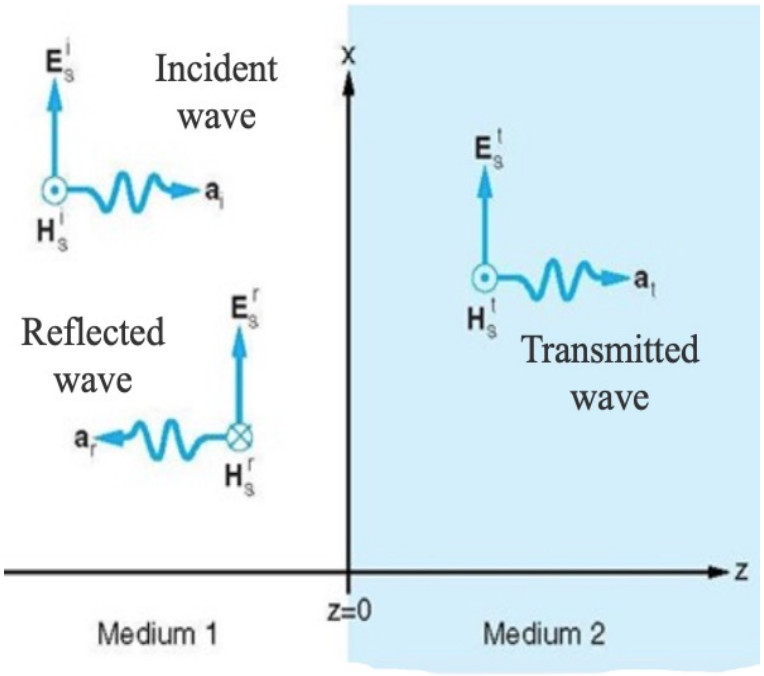
Electromagnetic Wave Optics

Part II: Light Behavior and Properties

The following subjects will be covered

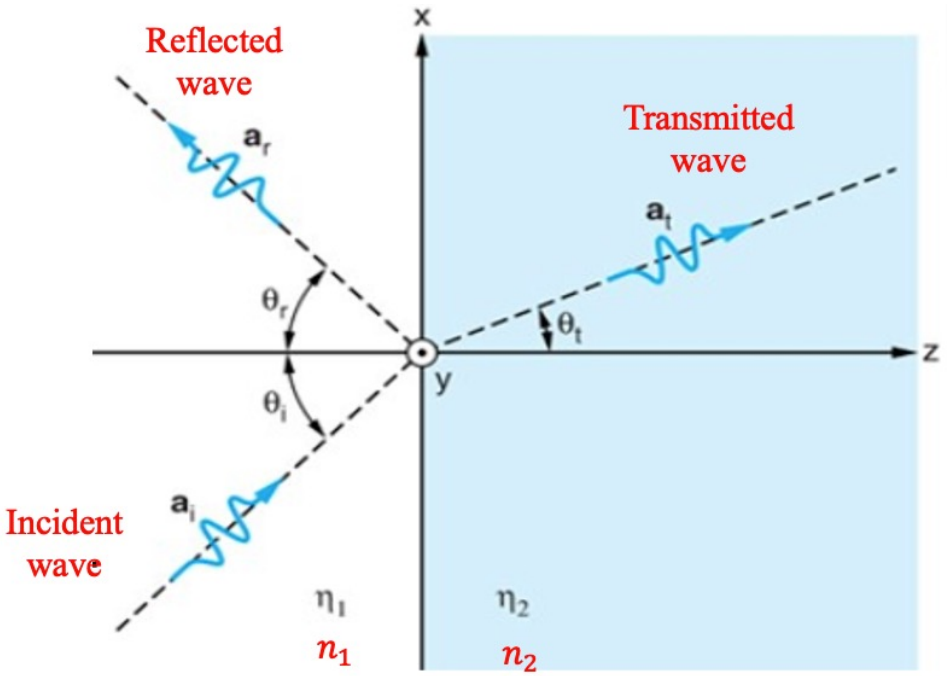
- Reflection and Transmission of Light.
- Light Polarization.
- Light-matter Interaction.
- Scattering of Light.

Light Reflection and Transmission



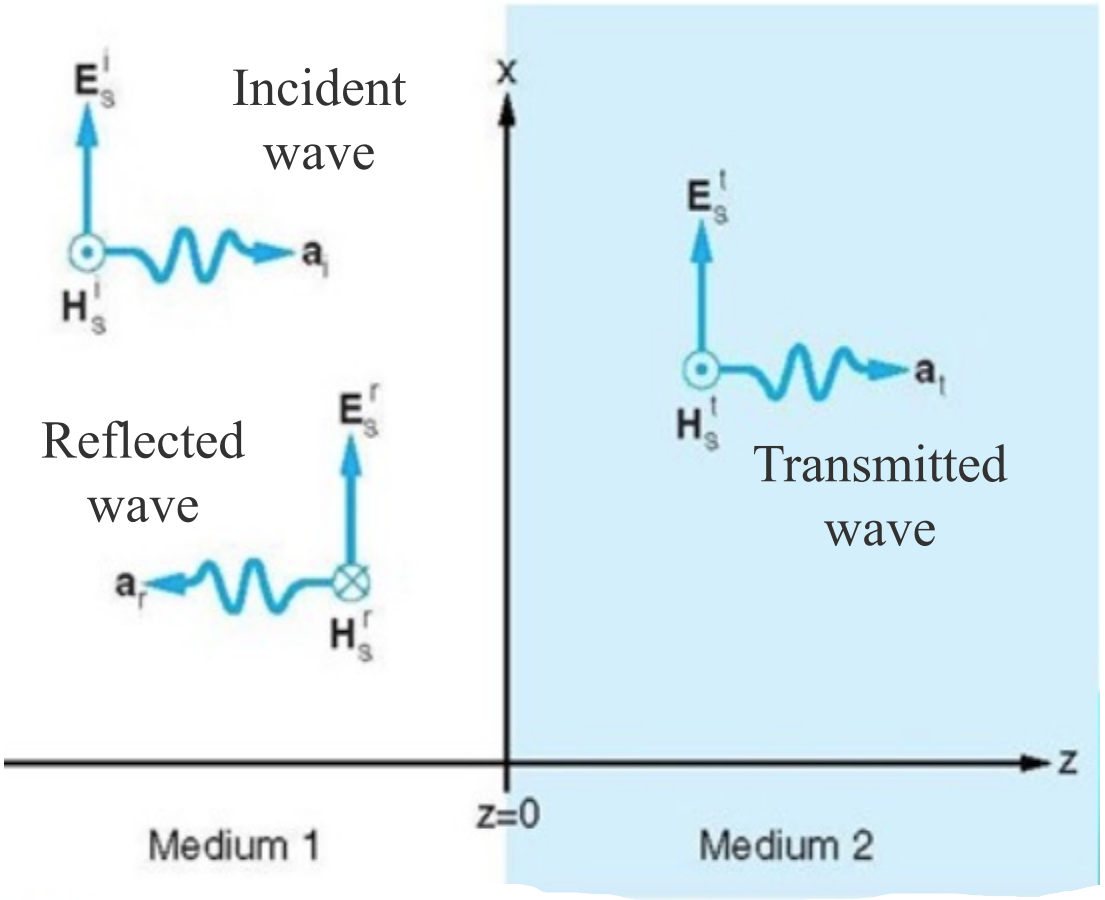
n_1, k_1, α_1 n_2, k_2, α_2

Normal Incidence



Oblique Incidence

Single Interface: Normal Incidence



n_1, k_1, α_1 n_2, k_2, α_2

Single Interface: Normal Incidence

Incident Wave

$$\vec{E}_i(z) = E_o\, e^{-jk_1 z}\, \vec{a}_x$$

$$\vec{H}_i(z) = \frac{E_o}{\eta_1} e^{-jk_1 z}\, \vec{a}_y$$

Reflected Wave

$$\vec{E}_r(z) = \Gamma\, E_o\, e^{+jk_1 z}\, \vec{a}_x$$

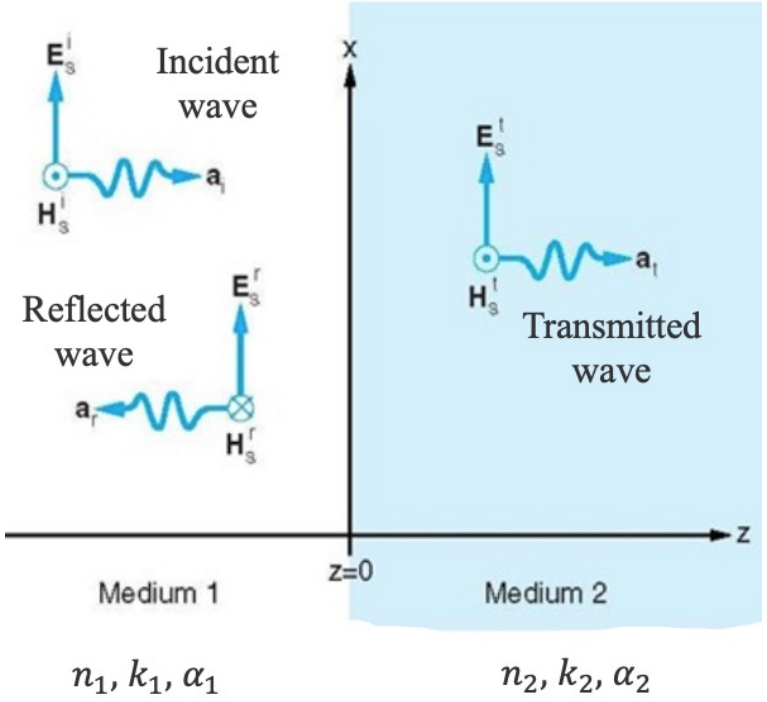
$$\vec{H}_r(z) = \frac{\Gamma E_o}{\eta_1}\, e^{+jk_1 z}\ \ (-\vec{a}_y)$$

Transmitted Wave

$$\vec{E}_t(z) = \tau E_o\, e^{-jk_2 z}\ \ \vec{a}_x$$

$$\vec{H}_t(z) = \frac{\tau\, E_o}{\eta_2}\, e^{-jk_2 z}\ \ \overrightarrow{a_y}$$

$$\eta_1 = \frac{\eta_o}{n_1} \qquad \eta_2 = \frac{\eta_o}{n_2} \qquad \eta_o = \sqrt{\mu_o/\varepsilon_o}$$



n_1, k_1, α_1 $\qquad$ n_2, k_2, α_2

Apply boundary conditions at interface $z = 0$

$$E_{tangential\ (1)} = E_{tangential\ (2)}$$

$$H_{tangential\ (1)} = H_{tangential\ (2)}$$

Single Interface: Normal Incidence

Apply boundary conditions

$$E_{1_tan} = E_o(1+\Gamma)$$

$$E_{2_tan} = E_o\tau$$

$$H_{1_tan} = \frac{E_o}{\eta_1}(1-\Gamma)$$

$$H_{2_tan} = \frac{\tau E_o}{\eta_2}$$

Field Coefficients

Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

$$\Gamma = \tau - 1$$

Transmission coefficient

$$\tau = \frac{2n_2}{n_1 + n_2}$$

Power Coefficients

Reflectance

$$R = |\Gamma|^2$$

$$R + T = 1$$

Transmittance

$$T = \frac{n_2}{n_1}|\tau|^2$$

Power Conservation

1) Γ and τ are in general complex.

2) $0 \le |\Gamma| \le 1$

$$\vec{S}^i_{ave} = \frac{n_1|E_o|^2}{2\eta_o}\,\vec{a}_k$$

Incident power density

$$\vec{S}^r_{ave} = n_1 R\frac{|E_o|^2}{2\eta_o}\,(-\vec{a}_k)$$

Reflected power density

$$\vec{S}^t_{ave} = n_1 T\frac{|E_o|^2}{2\eta_o}\,\vec{a}_k$$

Transmitted power density

Single Interface: Normal Incidence

Reflectance (Power)

$$R = |\Gamma|^2 = \left|\frac{n_1 - n_2}{n_1 + n_2}\right|^2$$

Transmittance (Power)

$$T = 1 - R = \left|\frac{4\, n_1 n_2}{(n_1 + n_2)^2}\right|$$

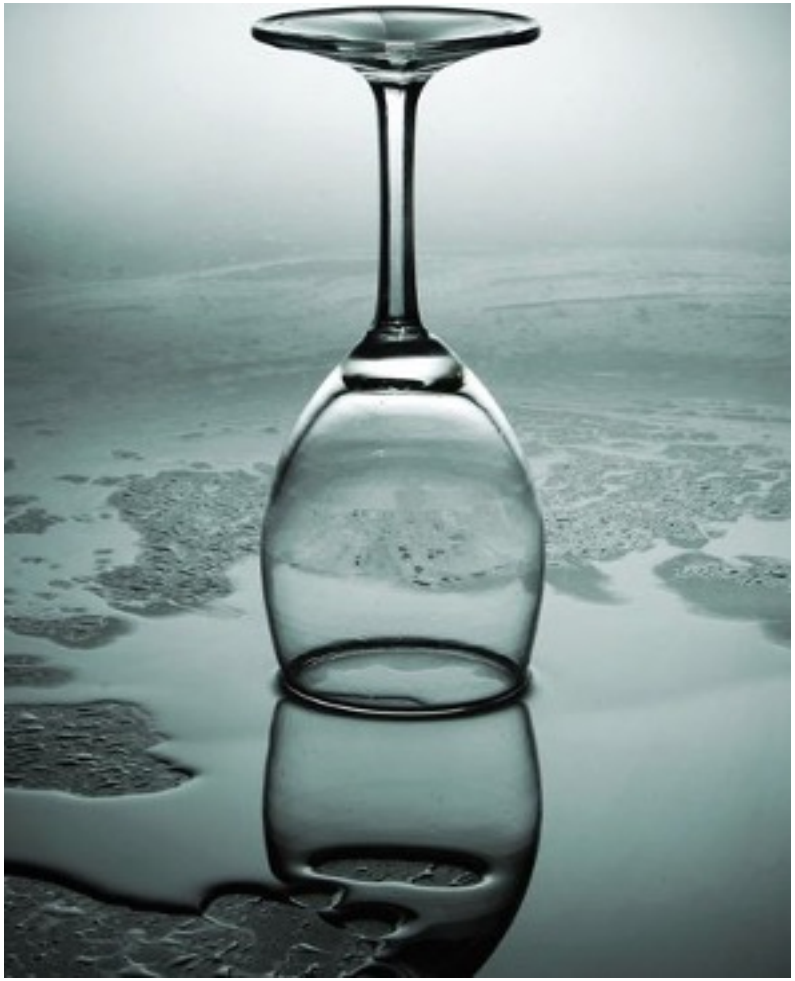
Example: Glass Reflection

Glass refractive index, $n \approx 1.52$

$$\Gamma = \frac{1.0-1.52}{1.0+1.52} = -\,0.2065$$

$$R = (-0.2065)^2 = 0.0426$$

$$T = 1 - 0.0426 = 0.957$$



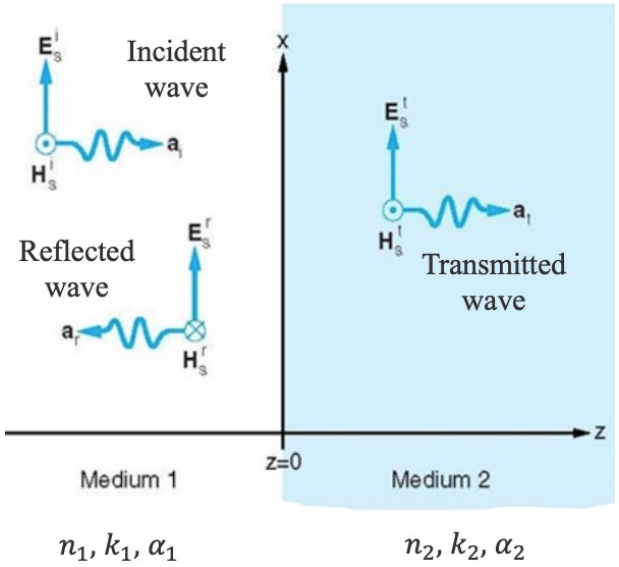
96% of light power goes through!

Remarks

1) Medium 1 perfect dielectric (n_1=real) and medium 2 perfect conductor ($n_2 \to \infty$)

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2} = -1$$ Total reflection

$$\tau = 0$$ Zero transmission



2) Medium 1 is a denser dielectric than medium 2

$$n_1 > n_2 \qquad \Gamma > 0$$

3) Medium 1 is a lighter dielectric than medium 2

$$n_1 < n_2 \qquad \Gamma < 0$$

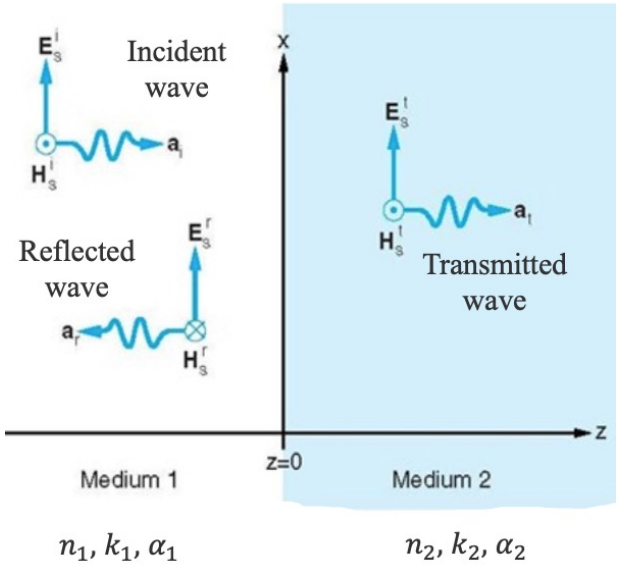
Lossy Media

Electric Fields

$$\overrightarrow{E_i}(z) = E_o\, e^{-\alpha_1 z}\, e^{-jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_r}(z) = \Gamma\, E_o\, e^{+\alpha_1 z}\, e^{+jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_t}(z) = \tau E_o\, e^{-\alpha_2 z}\, e^{-jk_2 z}\, \vec{a}_x$$



Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

Transmission coefficient

$$\tau = \frac{2n_2}{n_1 + n_2}$$

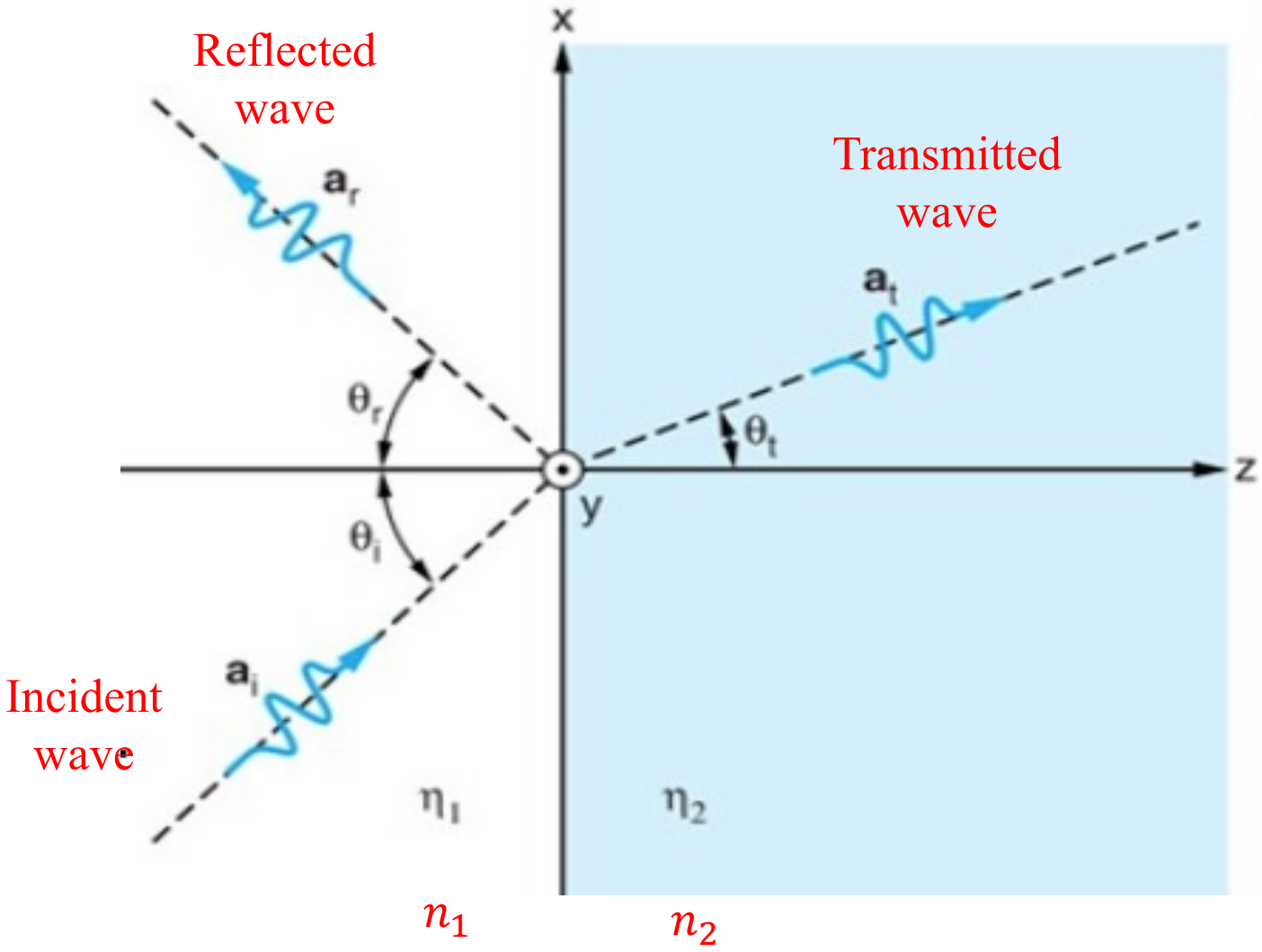
(n_1 and n_2 complex quantities)

$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2}\, e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, \vec{a}_z$$

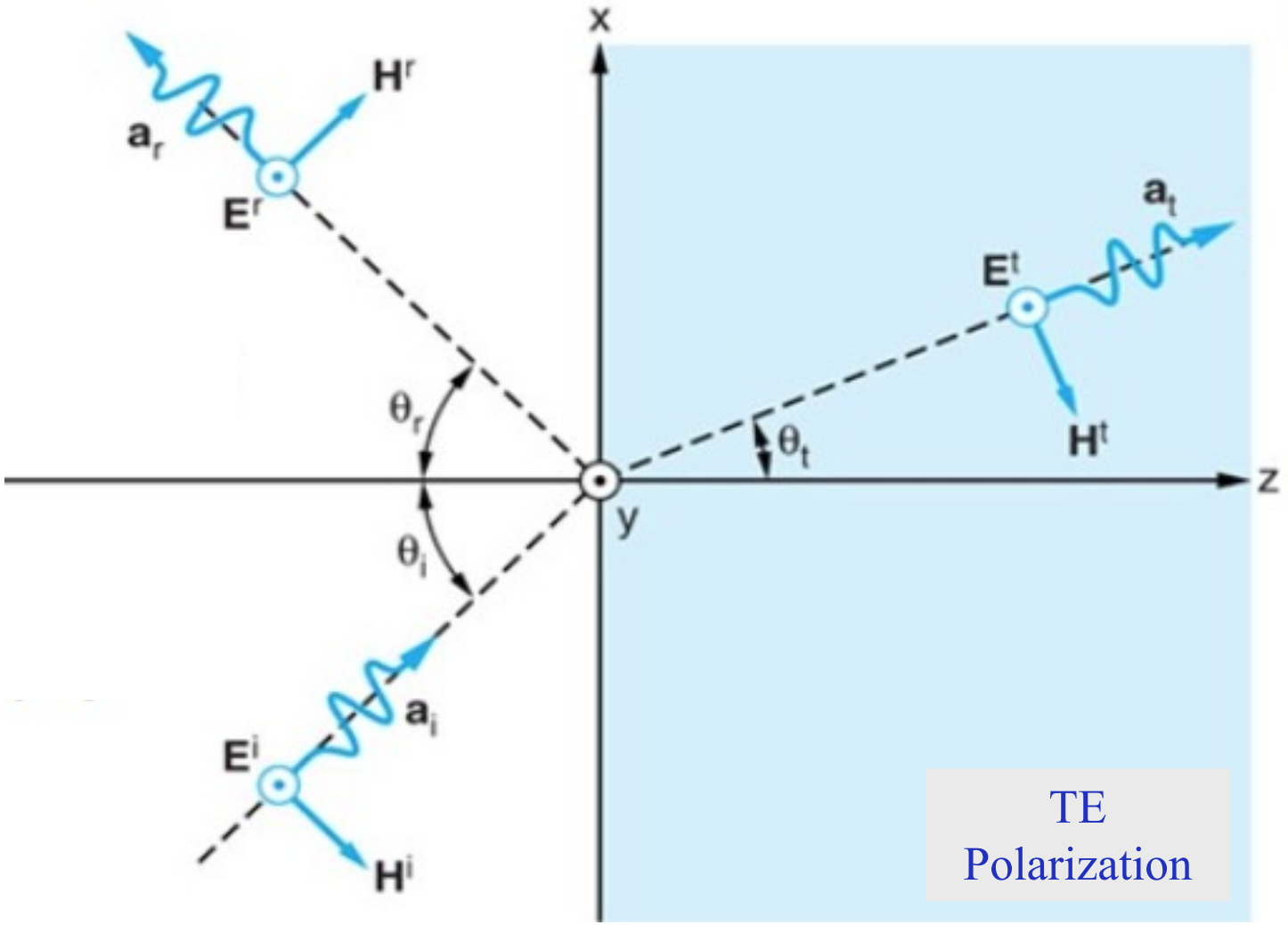
$$\vec{S}^r_{ave} = |\Gamma|^2\, \frac{|E_o|^2}{2}\, e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, (-\vec{a}_z)$$

$$\vec{S}^t_{ave} = |\tau|^2\, \frac{|E_o|^2}{2}\, e^{-2\alpha_2 z}\, Re\left(\frac{1}{\eta_2}\right)\, \vec{a}_z$$

Light Reflection and Transmission

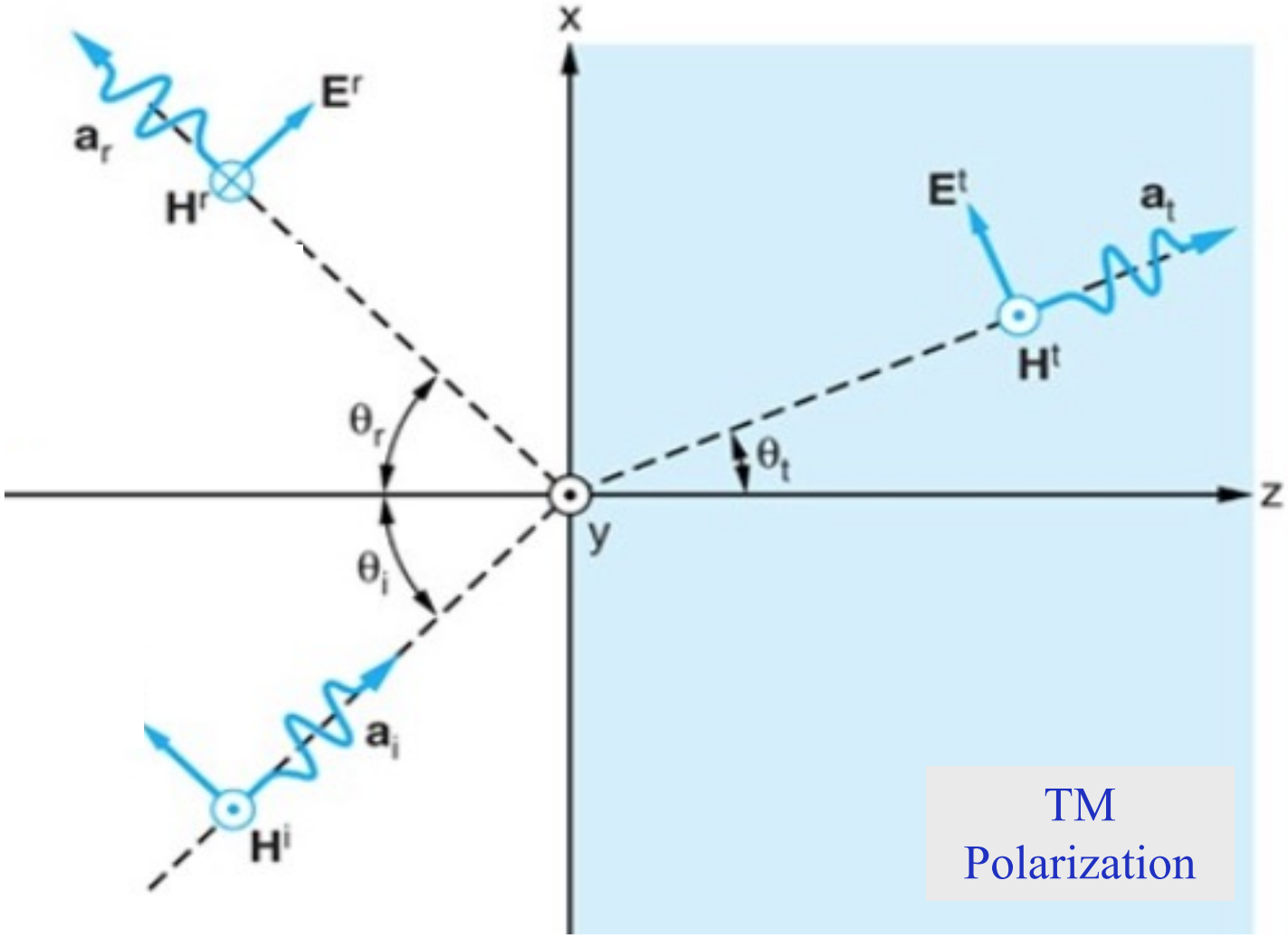


Single Interface: Oblique Incidence

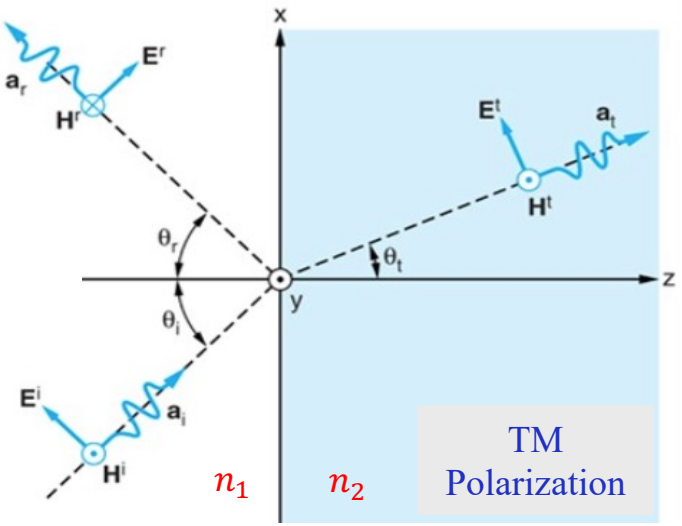


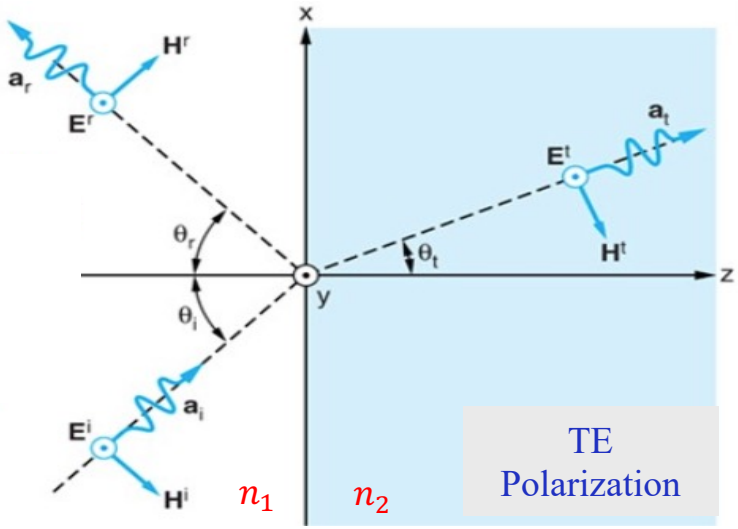
n_1 n_2

Single Interface: Oblique Incidence



Single Interface: Oblique Incidence





$$\Gamma_{TM} = \frac{n_1\,cos\theta_t - n_2\,cos\theta_i}{n_1\,cos\theta_t + n_2\,cos\theta_i}$$

$$\Gamma_{TM} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}$$

$$\tau_{TM} = \frac{2\,n_1\,cos\theta_i}{n_1\,cos\theta_t + n_2\,cos\theta_i}$$

$$\tau_{TM} = (1 + \Gamma_{TM})\frac{cos\theta_i}{cos\theta_t}$$

Snell's Law

$$sin\theta_t = \frac{n_1}{n_2}\,sin\theta_i$$

$$R = |\Gamma|^2$$

$$T = 1 - R$$

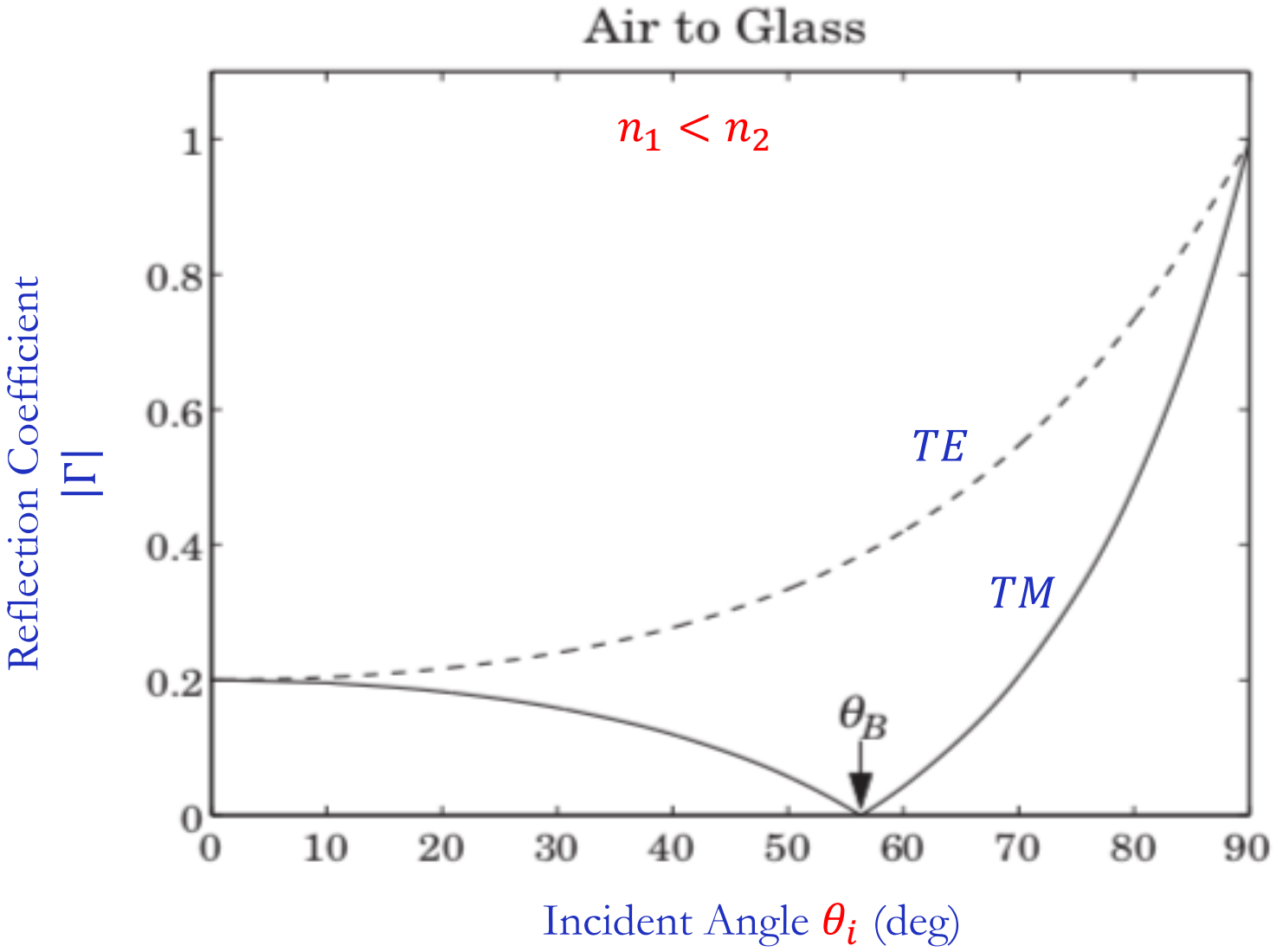
$$\Gamma_{TE} = \frac{n_1\,cos\theta_i - n_2\,cos\theta_t}{n_1\,cos\theta_i + n_2\,cos\theta_t}$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

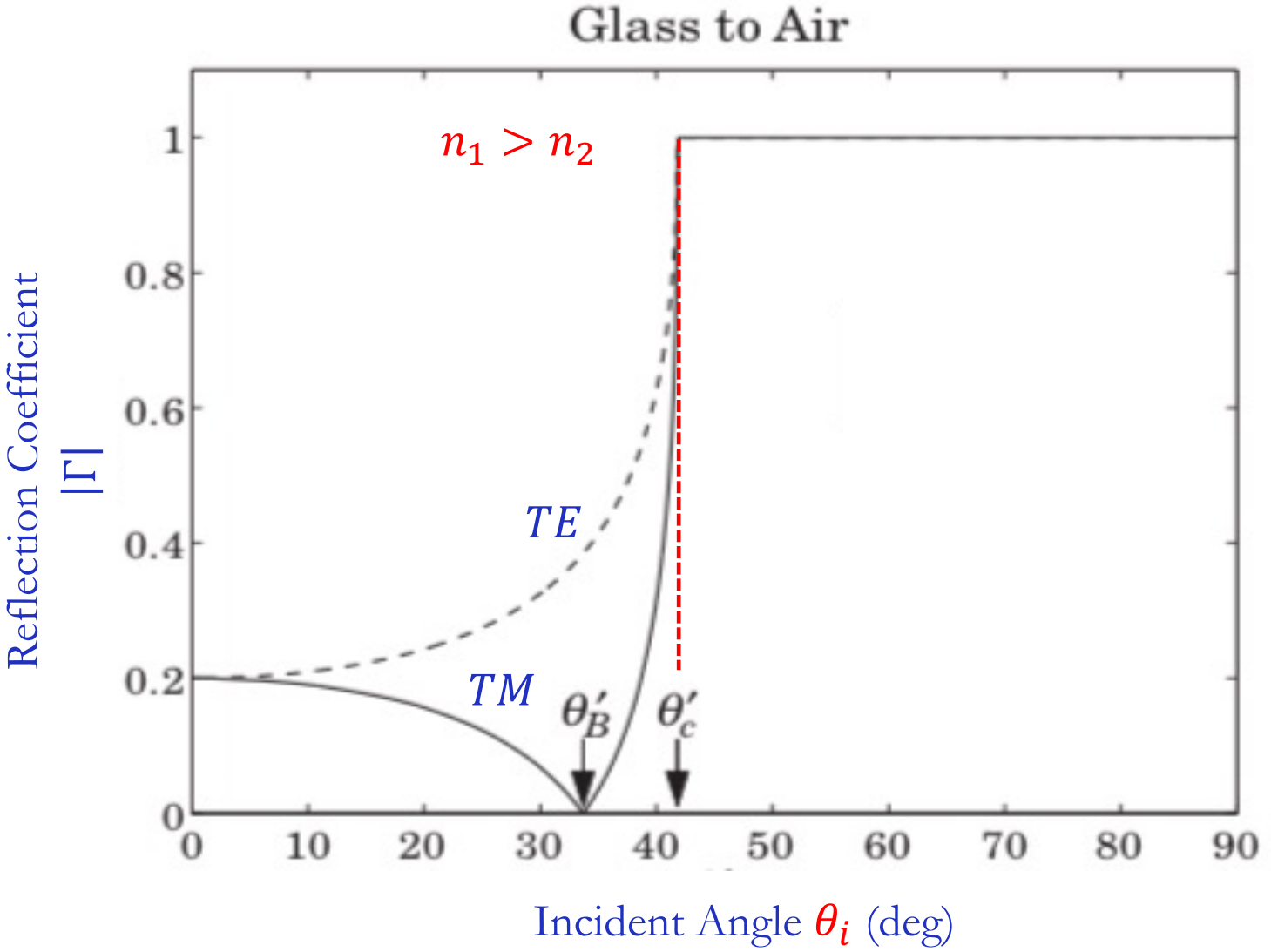
$$\tau_{TE} = \frac{2\,n_1\,cos\theta_i}{n_1\,cos\theta_i + n_2\,cos\theta_t}$$

$$\tau_{TE} = 1 + \Gamma_{TE}$$

Single Interface: Oblique Incidence



Single Interface: Oblique Incidence



Single Interface: Oblique Incidence

1) Total Transmission. $|\Gamma| = 0$, Only TM polarization.

Incident Angle $\theta_i = \tan^{-1}(n_2/n_1)$ Brewster Angle θ_B

In this case $\theta_i + \theta_t = 90^o$

Air-Glass: $\theta_B = \tan^{-1}(1.52/1)$

$\theta_B = 56.7^o$

2) Total Reflection $|\Gamma| = 1$, Both TE and TM polarizations, for $n_1 > n_2$ only.

Incident Angle $\theta_i = \sin^{-1}(n_2/n_1)$ Critical Angle θ_c

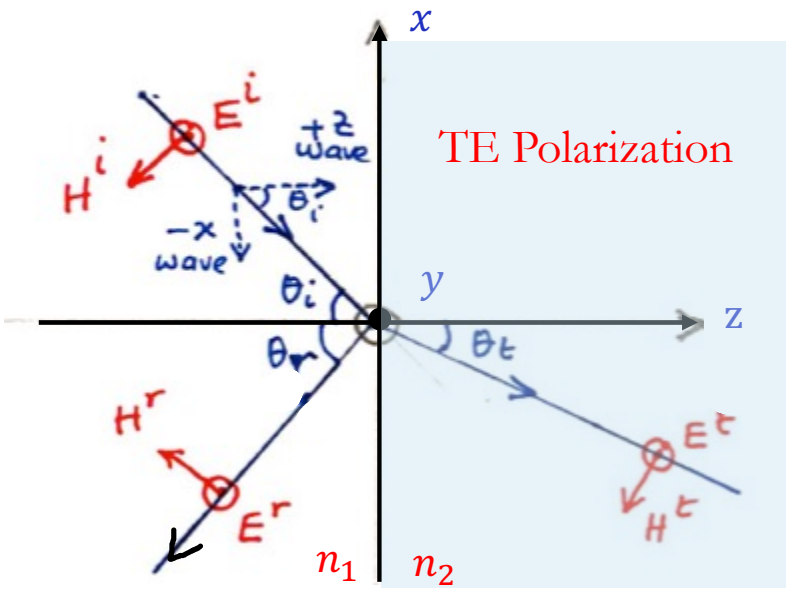
Happens for any angle $\theta_i \geq \theta_c$

Glass-Air: $\theta_c = \sin^{-1}(1/1.52)$

$\theta_c = 41.1^o$

Examples

A single-frequency plane wave is incident obliquely from air onto a planar interface with a non-magnetic medium having a refractive index of $\sqrt{2}$. The electric field of the incident wave has the following phasor form $\vec{E} = [\frac{1}{\sqrt{2}}\vec{a}_y]\, e^{-jk_o(z-\sqrt{2}x)/\sqrt{3}}$ V/m



From the $\vec{E}_i$ equation, the z incident wave takes the cosine. $\cos\theta_i = \frac{1}{\sqrt{3}}$

The x incident wave takes the sine. $\sin\theta_i = \frac{\sqrt{2}}{\sqrt{3}}$

$$\theta_i = 54.7°$$

From Snell's law, $\sin\theta_t = \frac{n_2}{n_1}\sin\theta_i$

$$\theta_t = 35.3°$$

$$n_1 \sin\theta_i = n_2 \sin\theta_t$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} = -0.333$$

$$\tau_{TE} = 1 + \Gamma_{TE} = 0.667$$

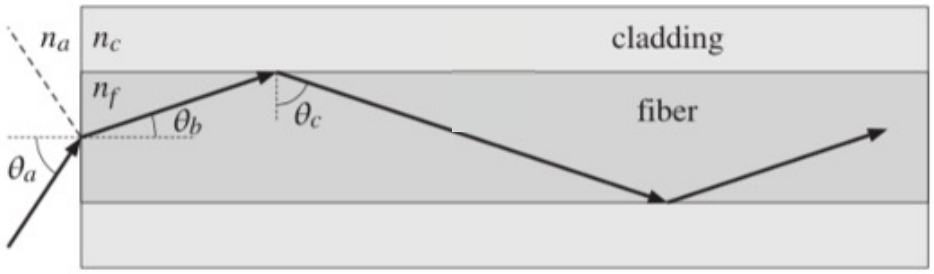
$$\vec{H}_r = \frac{(-0.333)}{(377)(\sqrt{2})}\, e^{jk_o(\sqrt{2}x+z)/\sqrt{3}}\, (\vec{a}_x - \sqrt{2}\vec{a}_z)/\sqrt{3} \text{ A/m}$$

$$\vec{E}_t = \frac{(0.667)}{(\sqrt{2})}\, e^{-j\sqrt{2}k_o(-x+\sqrt{2}z)/\sqrt{3}}\, (\vec{a}_y) \text{ V/m}$$

Examples

The Optical Fiber.

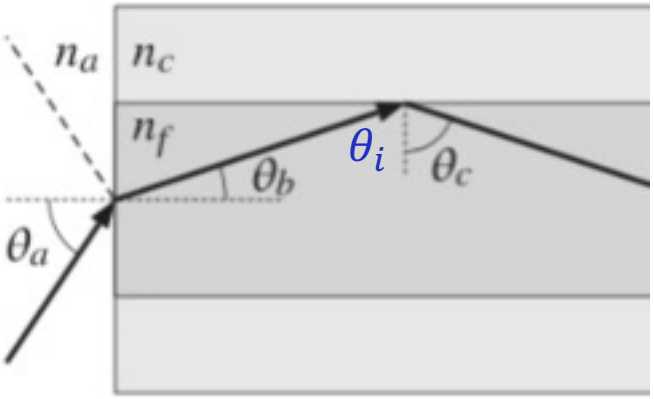
The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface. $\theta_i = \theta_c = 90 - \theta_b$



Snell's Law $n_a \sin\theta_a = n_f \sin\theta_b$

$$\sin\theta_a = \frac{n_f}{n_a}\sin\theta_b = \frac{n_f}{n_a}\sin(90-\theta_c)$$

$$\sin\theta_a = \frac{n_f}{n_a}\cos\theta_c = \frac{n_f}{n_a}\sqrt{1-(\sin\theta_c)^2}$$



Critical angle condition

$$\sin\theta_c = \frac{n_c}{n_f} \Rightarrow \sin\theta_a = \frac{\sqrt{(n_f)^2-(n_c)^2}}{n_a}$$

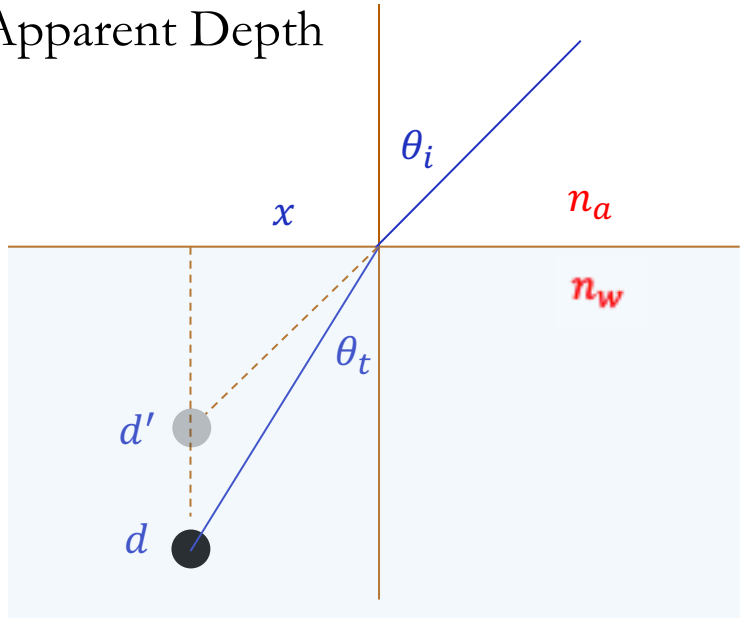
$n_a = 1, n_f = 1.49, n_c = 1.48$

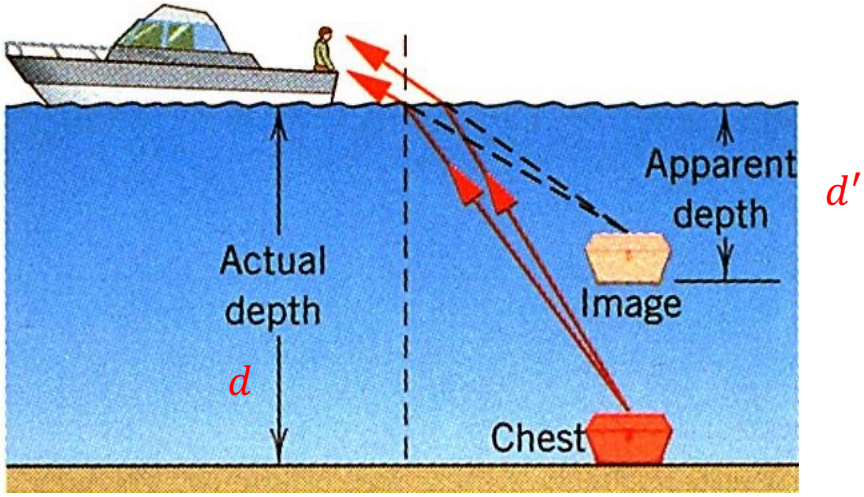
$\theta_c = 83.4°$

$\theta_a = 9.9°$

Examples

Apparent Depth





$$d = x \cot\theta_t$$

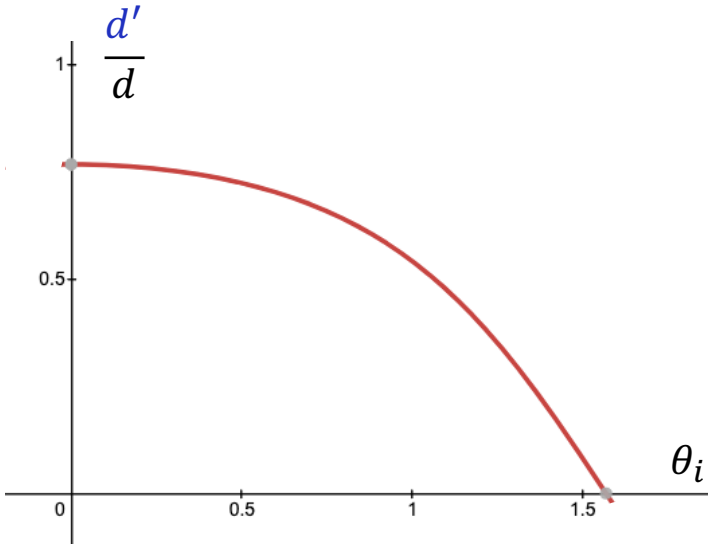
$$d' = x \cot\theta_i$$

$$n_a \sin\theta_i = n_w \sin\theta_t$$

$$\frac{d'}{d} = \frac{\cot\theta_i}{\cot\theta_t} = \frac{\cos\theta_i}{\cos\theta_t} \cdot \frac{\sin\theta_t}{\sin\theta_i}$$

$$\frac{d'}{d} = \frac{n_a}{n_w} \cdot \frac{\cos\theta_i}{\cos\theta_t}$$

$$\frac{d'}{d} = \frac{\cos\theta_i}{\sqrt{n_w^2 - (\sin\theta_i)^2}}$$



Q & A

Send me an email if you have any questions.